FR-03 – Step 2: Identify equivalence classes for each input & output

Input: the variables from `FR-03_1_variables.md`. Method: lecture S04 *Domain Testing*, step 2.

Guidelines applied (S04, slides 8–11)

Condition kind	Rule	Example in this feature
Range	1 valid class + 2 invalid classes (below, above)	OTP length = 6, password length ≥ 8, OTP age < T
Set	1 valid class per element (if handled differently) + 1 invalid class	"Back to login" step, confirm field in API
Must be	1 valid class + 1 invalid class	Email registered, OTP digits only, password has uppercase
Split	Split a class when its members may not be handled the same way	Email without TLD, OTP with leading zero, confirm differing only in case

Legend: V = valid, I = invalid, TBD = expected result decided during execution. "Tester decision" = the spec is silent; the expected result was chosen by the tester (see the decision log). Representative values are **not** chosen here; that is step 3.

Input equivalence classes

FR-03_IN_01 – Email (step 1)

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-03_EC_01	C1 Required (must be)	Email is not empty	V	Assumption
FR-03_EC_02	C1 Required (must be)	Email is empty or whitespace only	I	Assumption
FR-03_EC_03	C2 Format `user@domain.com` (must be)	Email is well-formed	V	Assumption (FR-01)
FR-03_EC_04	C2 Format (split)	No `@`	I	Assumption (FR-01)
FR-03_EC_05	C2 Format (split)	Empty local part (nothing before `@`)	I	Assumption (FR-01)
FR-03_EC_06	C2 Format (split)	No top-level domain (`user@domain`); browser `type="email"` accepts this, spec format does not	I	Assumption (FR-01)
FR-03_EC_07	C3 Registered (must be)	Registered email, typed exactly as stored	V	Spec
FR-03_EC_08	C3 Registered (must be)	Well-formed but not registered	I	Spec
FR-03_EC_09	C3 Registered (split)	Registered email in different letter case	I	Tester decision: case-sensitive
FR-03_EC_10	C3 Registered (split)	Registered email with leading/trailing spaces	I	Tester decision: not trimmed

FR-03_IN_02 – "Back to login" button

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-03_EC_11	C1 Where clicked (set)	Clicked on step 1	V	Spec
FR-03_EC_12	C1 Where clicked (set)	Clicked on step 2 (assumes the button is also there)	V	Assumption

No invalid class: a button has no out-of-set value.

FR-03_IN_03 – OTP (step 2)

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-03_EC_13	C1 Required (must be)	OTP is not empty	V	Assumption
FR-03_EC_14	C1 Required (must be)	OTP is empty	I	Assumption
FR-03_EC_15	C2 Length = 6 (range)	Length = 6	V	Spec
FR-03_EC_16	C2 Length = 6 (range)	Length < 6	I	Spec
FR-03_EC_17	C2 Length = 6 (range)	Length > 6	I	Spec
FR-03_EC_18	C3 Digits only (must be)	All characters are digits 0–9	V	Spec
FR-03_EC_19	C3 Digits only (must be)	Contains a letter or symbol	I	Spec
FR-03_EC_20	C3 Digits only (split)	6 digits with leading/trailing spaces	V	Tester decision: trimmed
FR-03_EC_21	C4 Matches issued OTP (must be)	Equals the OTP issued for this email	V	Spec
FR-03_EC_22	C4 Matches issued OTP (must be)	6 digits but not the issued OTP	I	Spec
FR-03_EC_23	C4 Matches issued OTP (split)	Correct OTP that starts with `0` (risk: treated as a number, zero dropped)	V	Spec

FR-03_IN_04 – New password

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-03_EC_24	C1 Required (must be)	Not empty	V	Assumption
FR-03_EC_25	C1 Required (must be)	Empty	I	Assumption
FR-03_EC_26	C2 Length ≥ 8 (range)	Length ≥ 8	V	Spec (FR-01)
FR-03_EC_27	C2 Length ≥ 8 (range)	Length < 8	I	Spec (FR-01)
FR-03_EC_28	C2 Length (range, upper side)	Very long (e.g. > 128 characters); no maximum in spec	I	Tester decision: too long is rejected
FR-03_EC_29	C3 Uppercase (must be)	Has ≥ 1 uppercase letter	V	Spec (FR-01)
FR-03_EC_30	C3 Uppercase (must be)	No uppercase letter	I	Spec (FR-01)
FR-03_EC_31	C4 Lowercase (must be)	Has ≥ 1 lowercase letter	V	Spec (FR-01)
FR-03_EC_32	C4 Lowercase (must be)	No lowercase letter	I	Spec (FR-01)
FR-03_EC_33	C5 Digit (must be)	Has ≥ 1 digit	V	Spec (FR-01)

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-03_EC_34	C5 Digit (must be)	No digit	I	Spec (FR-01)
FR-03_EC_35	C6 Special from `@$!%*?&` (must be)	Has ≥ 1 special character from the set	V	Spec (FR-01)
FR-03_EC_36	C6 Special (must be)	No special character at all	I	Spec (FR-01)
FR-03_EC_37	C6 Special (split)	Special characters only from outside the set (`#`, `_`, `-`, `.`)	I	Spec (strict reading)
FR-03_EC_38	C7 Other characters (split)	Contains a space or non-ASCII letters (e.g. Vietnamese diacritics)	I	Tester decision: not allowed
FR-03_EC_39	C8 Differs from current (split)	Same as the current password	V	Tester decision: allowed

The 7 allowed special characters are kept as **one** valid class: there is no reason in the spec to expect them to be handled differently.

FR-03_IN_05 – Confirm new password

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-03_EC_40	C1 Equals new password (must be)	Identical to new password	V	Spec
FR-03_EC_41	C1 (must be)	Empty	I	Assumption
FR-03_EC_42	C1 (split)	Completely different	I	Spec
FR-03_EC_43	C1 (split)	Differs only in letter case	I	Spec
FR-03_EC_44	C1 (split)	Differs only by a trailing space	I	Spec

FR-03_IN_06 – OTP age

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-03_EC_45	C1 Age < T (range)	0 ≤ age < T	V	Spec (SEC-07) + Assumption (T)
FR-03_EC_46	C1 Age < T (range)	Age ≥ T (expired)	I	Spec (SEC-07) + Assumption (T)

Only one invalid class: age cannot be negative.

FR-03_IN_07 – OTP use count

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-03_EC_47	C1 Unused (must be)	Never used (count = 0)	V	Spec (SEC-07)
FR-03_EC_48	C1 Unused (must be)	Already used in a successful reset (count ≥ 1)	I	Spec (SEC-07)
FR-03_EC_49	C1 Unused (split)	Submitted before, but that reset failed validation (e.g. weak password)	I	Tester decision: failed reset consumes OTP

FR-03_IN_08 – Email sent with the reset request (API)

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-03_EC_50	C1 Same as requesting email (must be)	Same email that requested the OTP	V	Spec (rule)
FR-03_EC_51	C1 (split)	A different, registered email	I	Spec (rule)
FR-03_EC_52	C1 (split)	An unregistered email	I	Assumption
FR-03_EC_53	C1 (split)	Missing or empty	I	Assumption

FR-03_IN_09 – Number of OTP requests for the same email

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-03_EC_54	C1 Latest OTP only (set)	Only one OTP requested; use it	V	Assumption
FR-03_EC_55	C1 (set)	Several requested; use the latest	V	Assumption
FR-03_EC_56	C1 (set)	Several requested; use an older one	I	Assumption

FR-03_IN_10 – Confirm-password field in the reset API request

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-03_EC_57	C1 Field state (set)	Present and matching	V	Assumption
FR-03_EC_58	C1 (set)	Present and not matching	I	Assumption
FR-03_EC_59	C1 (set)	Absent (as in API spec 1.4)	TBD	Decide after execution

FR-03_IN_11 – Step 1 completed before step 2

Class ID	Condition (guideline)	Equivalence class	V / I	Source
FR-03_EC_60	C1 Step order (must be)	Step 1 completed, then step 2	V	Assumption
FR-03_EC_61	C1 Step order (must be)	Step 2 submitted for an email that never requested an OTP	I	Assumption

Output equivalence classes

As in the lecture's SUM / Error example, each distinct output is one class. "Triggered by" links each output to the input classes expected to produce it; step 3 uses this to pick test cases.

Class ID	Output variable	Equivalence class	Triggered by	Source
FR-03_EC_62	OUT_01 Step indicator	"Bước 1 / 2" shown on step 1	Opening the page	Spec
FR-03_EC_63	OUT_01 Step indicator	"Bước 2 / 2" shown on step 2	EC_01 + EC_03 + EC_07	Spec
FR-03_EC_64	OUT_02 Generated OTP	6-digit OTP displayed	EC_01 + EC_03 + EC_07	Spec
FR-03_EC_65	OUT_03 Error: email not registered	Message shown	EC_08, EC_09	Assumption
FR-03_EC_66	OUT_04 Error: invalid email format	Message shown	EC_04, EC_05, EC_06, EC_10	Assumption

Class ID	Output variable	Equivalence class	Triggered by	Source
FR-03_EC_67	OUT_05 Error: required field	Message shown	EC_02, EC_14, EC_25, EC_41	Assumption
FR-03_EC_68	OUT_06 Error: incorrect OTP	Message shown	EC_16, EC_17, EC_19, EC_22, EC_56	Assumption
FR-03_EC_69	OUT_07 Error: OTP expired	Message shown	EC_46	Spec (SEC-07)
FR-03_EC_70	OUT_08 Error: OTP already used	Message shown	EC_48, EC_49	Spec (SEC-07)
FR-03_EC_71	OUT_09 Error: OTP not valid for this email	Message shown	EC_51, EC_52, EC_53, EC_61	Spec
FR-03_EC_72	OUT_10 Error: weak password	Message shown	EC_27, EC_28, EC_30, EC_32, EC_34, EC_36, EC_37, EC_38	Spec (FR-01)
FR-03_EC_73	OUT_11 Error: passwords don't match	Message shown	EC_42, EC_43, EC_44, EC_58, (EC_59 if the server enforces matching)	Spec
FR-03_EC_74	OUT_12 Success message	Message shown	All step 2 inputs in valid classes	Assumption
FR-03_EC_75	OUT_13 Redirect after success	Redirected to login	Same as EC_74	Assumption
FR-03_EC_76	OUT_14 Redirect via "Back to login"	Redirected to login	EC_11, EC_12	Spec
FR-03_EC_77	OUT_15 Account password	Changed: new password logs in, old one is rejected	Same as EC_74	Assumption
FR-03_EC_78	OUT_15 Account password	Unchanged: old password still logs in	Any invalid step 2 class	Assumption

Summary

	Count
Valid input classes	26
Invalid input classes	34
TBD input classes (decided at execution)	1
Output classes	17
Total	**78**

Decision log (spec silent, decided by the tester)

Bugs raised from these classes must say they rest on a tester decision, not on the spec.

Class	Question	Decision	Expected output
EC_09	Is the email case-insensitive?	No: different case is **invalid**	OUT_03 (not registered)
EC_10	Are spaces around the email trimmed?	No: **invalid**	OUT_04 (invalid format)
EC_20	Are spaces around the OTP trimmed?	Yes: **valid**	Proceeds as a correct OTP

Class	Question	Decision	Expected output
EC_28	Very long password (> 128 characters)	**Invalid**: must be rejected	OUT_10 (password rule error)
EC_38	Spaces / non-ASCII letters in password	**Invalid**	OUT_10 (password rule error)
EC_39	New password equal to current one	**Valid**	OUT_12 (success)
EC_49	Does a failed reset consume the OTP?	Yes: the OTP is **no longer valid**	OUT_08 (already used)
EC_59	Reset API request without a confirm field	**Decided during execution** (step 5)	Recorded with the result